Supplementary Information
On the Conservation Laws and Pathway Realizability of Biochemical Reaction Networks

Thanh-An Pham[1] and Tieu-Long Phan[1,2,*]

[1]Bioinformatics Group, Department of Computer Science & Interdisciplinary Center for Bioinformatics & School for Embedded and Composite Artificial Intelligence (SECAI), Leipzig University, Härtelstraße 16–18, D-04107 Leipzig, Germany
[2]Department of Mathematics and Computer Science, University of Southern Denmark, DK-5230 Odense M, Denmark

*Corresponding author: tieu@bioinf.uni-leipzig.de
Thanh-An Pham: thanh-an.pham@uni-leipzig.de

S1 Python interface example

Code Example S1 illustrates network construction, exact structural analysis, and discrete pathway verification through `SynKit`'s Python interface. The script builds a 14-species glycolysis model (with currency metabolites omitted), evaluates chemical reaction network theory (CRNT) invariants and deficiency, isolates conserved pools and minimal siphons, and verifies that one glucose token can reach two pyruvates via an exact 15-step firing sequence. Run this example from the `SynKit` source tree or unpacked source distribution: the research modules under `Experiment/CRN` are distributed with the source, separately from the installed runtime wheel.

```python
from synkit.CRN import (build_native, native_backend, crnt_summary,
    conserved_moieties, find_siphons, siphon_persistence_details,
    PathwayRealizability, crn_to_sbml)
from Experiment.CRN.studies import build_kegg_crn, GLYCOLYSIS_FLUX

library = build_native("/tmp/synkit-crn", with_semiflows=True)
with native_backend(library):
    crn = build_kegg_crn("M00001", drop_currency=True)
    crn.n_species, crn.n_reactions
    report = crnt_summary(crn)
    report.deficiency, report.is_weakly_reversible
    len(conserved_moieties(crn)), len(find_siphons(crn))
    siphon_persistence_details(crn).persistence_ok

    check = PathwayRealizability().load_syncrn_and_flow(
        crn, flow=GLYCOLYSIS_FLUX,
        initial_marking={"alpha-D-Glucose": 1},
        species="label", reaction="label")
    check.build_petri_net_from_flow()
    outcome = check.realizability_result()
    outcome.status.value, len(outcome.certificate)
    sbml_document = crn_to_sbml(crn)
```

Code Example S1 Python analysis of reduced `KEGG` glycolysis (`M00001`): network construction, exact structural analysis, discrete pathway verification, and `SBML` export.

S2 Reference-network validation and numerical comparison

Table S1 details the literature sources, targeted structural validation properties, and conservation-law audit for the twelve reference networks.

Table S1 Reference reaction networks: literature sources, targeted structural validation properties, and conservation-law numerical audit. SVD residual reports max $|\mathbf{S}^T B|$ for a floating-point basis B, Rounded invalid counts columns of round$(1000B)$ with non-zero integer residuals, and Exact residual evaluates `SynKit`'s integer basis. The open reactor has a zero-dimensional conservation space.

Network model	Source	Targeted property	SVD res.	Rounded inv.	Exact res.
Reversible isomerization	Feinberg [1], Ch. 3	Minimal weakly reversible, Deficiency Zero Theorem (positive branch)	1.1e-16	0	0
Irreversible isomerization	Feinberg [1], Ch. 3	Deficiency Zero Theorem (negative branch)	0.0e+00	0	0
Isomerization cycle	Feinberg [1], Ch. 3	Weak reversibility without individual reversible steps	2.2e-16	0	0
Three disjoint reversible pairs	Constructed test network	Disconnected linkage classes, with minimal semiflow supports isolating disjoint pairs rather than their union	1.1e-16	0	0
Disjoint pairs, non-unit stoichiometry	Constructed test network	Non-unit stoichiometry $E \rightleftharpoons 2F$: primitive conserved pool $2E + F$	1.1e-16	0	0
Michaelis-Menten	Michaelis & Menten, Feinberg [1]	Enzyme mechanism with two conserved pools, $\delta = 0$, non-weakly reversible	2.2e-16	1	0
Reversible Michaelis-Menten	Feinberg [1]	$\delta = 0$, with weak reversibility restored	1.7e-16	1	0
Edelstein bistable switch	Edelstein [2]	Minimal reversible $\delta = 1$ network with multiple steady states	0.0e+00	0	0
Futile cycle (single-site)	Wang & Sontag [3]	Distributive cycle with three independent pools	3.3e-16	3	0
Futile cycle (two-site)	Wang & Sontag [3]	Exact conservation in a multistationarity motif	5.3e-16	0	0
Shinar-Feinberg ACR motif	Shinar & Feinberg [4]	ACR motif, where a minimal siphon contains no P-semiflow support	1.1e-16	0	0
Open reactor $(0 \to A \to B \to 0)$	Constructed test network	Zero-complex handling, where cycle closure changes weak reversibility	0.0e+00	0	0

S2.1 Algebraic consistency and deficiency validation

We cross-checked deficiency using two formulations. With species-by-complex matrix $\mathbf{Y}$ and directed complex-graph incidence $\mathbf{I}_a$, we have $\mathbf{S} = \mathbf{Y}\mathbf{I}_a$ and

$$\delta \;=\; n - \ell - \operatorname{rank}(\mathbf{S}) \;=\; \operatorname{rank}(\mathbf{I}_a) - \operatorname{rank}(\mathbf{Y}\mathbf{I}_a). \tag{1}$$

Both formulations agreed for every benchmark, confirming consistency among complex partitioning, rank computation, and stoichiometric assembly. Fig. S1 illustrates the cross-check and the theorem verdicts on the Edelstein network.

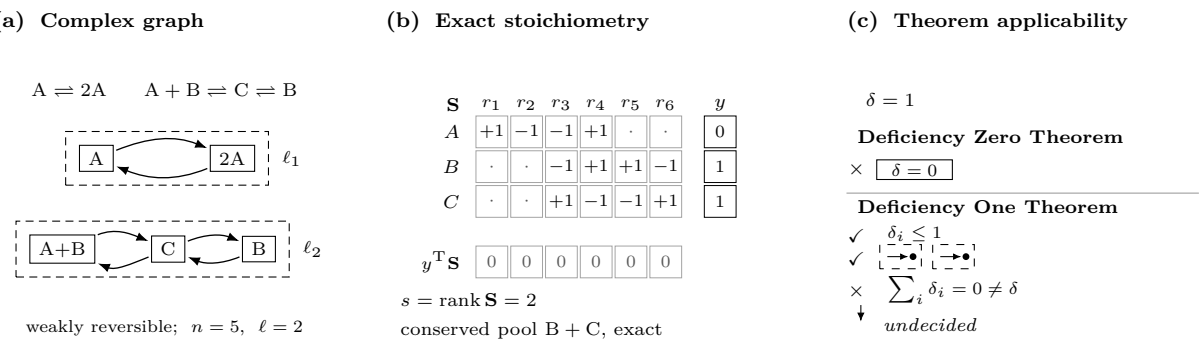


Figure S1 Conservation structure and deficiency-theorem checks for the Edelstein network [2]. **(a)** Complex graph and its two linkage classes. **(b)** Stoichiometric matrix beside the primitive conservation vector y, whose product $y^{\mathsf{T}}\mathbf{S}$ vanishes over $\mathbb{Q}$ and certifies the conserved pool [B] + [C]. **(c)** Each classical deficiency theorem fails one hypothesis, leaving multistationarity undecided by deficiency alone.

S2.2 Numerical nullspaces and exact integer conservation laws

We audit integer representatives y satisfying $y^T\mathbf{S} = 0$ on the twelve reference networks (Table S1). The floating-point basis B is computed by `scipy.linalg.null_space` applied to $\mathbf{S}^T$. Its maximum absolute residual ranges from zero to 5.3×10^{-16}; irreversible isomerization and the Edelstein network have zero reported residual. We separately evaluate the heuristic round$(10^3 B)$ on the integer matrix. It produces vectors with non-zero exact residuals in Michaelis–Menten, reversible Michaelis–Menten, and the one-site futile cycle. This audit demonstrates the effect of this particular scaling and rounding choice on integer identities. `SynKit`'s exact rational elimination returns a basis of the expected dimension for every network, including the empty basis for the open reactor; every returned integer vector satisfies the conservation identity exactly.

S3 Computational cost across network families

Fig. S2 illustrates the three benchmark families. The complete scaling report contains 60 measurements: three families, four requested sizes, and five tasks. Fig. S3 shows the 48 measurements for four structural analyses, with numbers beside the random-sparse siphon points giving the number of minimal siphons. The remaining twelve measurements are standalone exact-rank computations; the reversible-chain rank timings also appear in Fig. 3c.

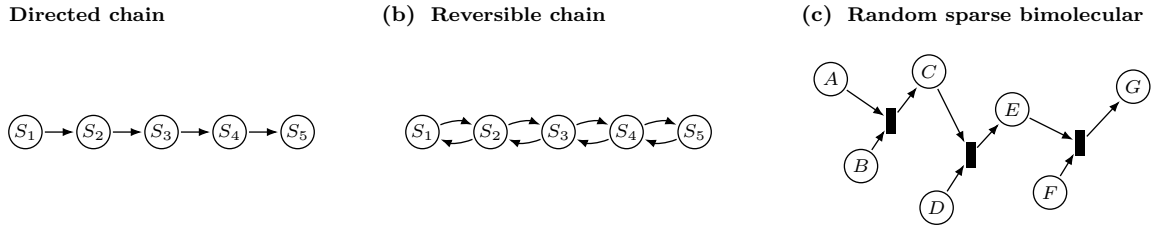


Figure S2 Network families used in the scaling benchmark. **(a)** Directed chain with one-way conversion steps. **(b)** Reversible chain with paired forward and reverse steps. **(c)** Random sparse bimolecular network with two substrates per reaction and sparse connectivity. Circles denote species and bars denote reactions.

S4 BioModels: structural SBML interoperability

The eleven entries in Table S2 benchmark structural `SBML` exchange across a size-stratified selection from `BioModels`: ten reaction-bearing models pass export and re-import checks without loss of integer stoichiometric multiplicities, while `BIOMD0000000562` serves as a reaction-free control. The packaged independent reference, `sbml_reference.py` in `Experiment/CRN/studies`, computes eight shared CRNT quantities directly from the `SBML` XML using only the Python standard library and exact rational arithmetic. The participant-conventions study compares this reference with the public importer for the retained, cancelled and modifier-added readings of every reaction-bearing model. Inputs are verified against the packaged SHA-256 manifest. Reversible reactions are expanded into distinct directed steps: an omitted flag means true in

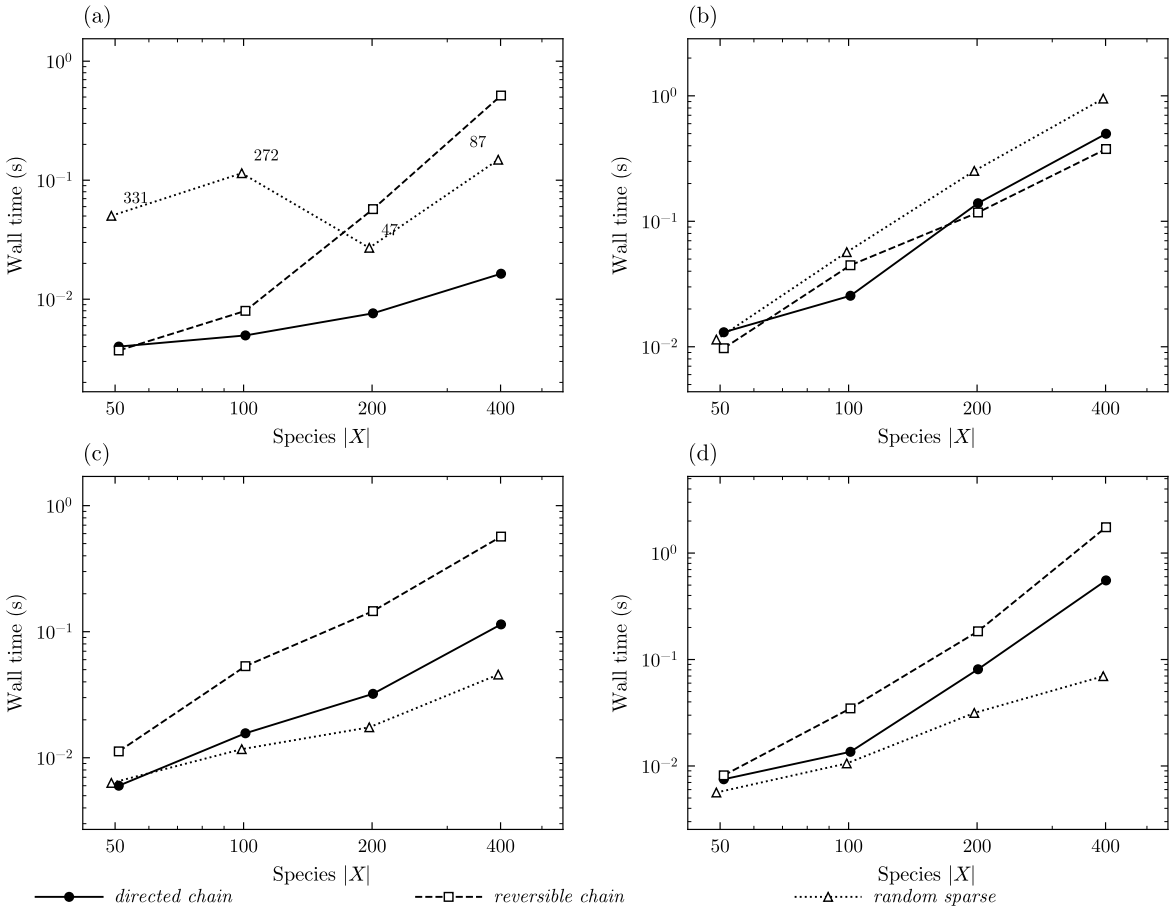


Figure S3 Structural analysis runtimes across directed-chain, reversible-chain, and random-sparse networks. **(a)** Minimal-siphon enumeration, where numbers beside random-sparse points give minimal siphon counts. **(b)** Exact CRNT analysis. **(c)** Graph canonicalization. **(d)** Exact conserved-pool enumeration. Points and connecting lines show 48 completed single-run measurements across these four tasks. Line styles and markers distinguish the three families. All 60 requested measurements, including twelve standalone exact-rank computations, complete.

Level 2; Level 3 requires an explicit flag, with a strict error or lenient warning when absent. The BioModels enumeration uses a 1 000 000-node budget; incomplete searches are flagged and excluded from completed counts and timings.

Table S2 BioModels input inventory and structural results from the version 1.7.0 source snapshot. m: species, r: directed reactions after reversibility expansion, s: exact rank, δ: deficiency, P: minimal conserved pools, S: minimal siphons, RT: supported structural round trip, and Covered: every minimal siphon contains a conserved-pool support. Level 2 omitted reversibility flags default to true. Coverage is the sufficient persistence condition used in the main text, including the vacuous case of no siphons; failure does not prove extinction. Dashes denote unavailable quantities or analyses inapplicable to the reaction-free control; completion status is recorded in the accompanying report.

Model	m	r	s	δ	P	S	RT	Covered
BIOMD0000000001	12	34	11	0	1	1	yes	yes
BIOMD0000000010	8	10	5	0	3	3	yes	yes
BIOMD0000000051	18	96	18	8	0	0	yes	yes
BIOMD0000000064	26	28	17	0	11	7	yes	no
BIOMD0000000108	9	18	7	8	2	2	yes	yes
BIOMD0000000404	29	60	19	13	10	10	yes	yes
BIOMD0000000637	12	35	9	9	3	3	yes	yes
BIOMD0000000019	100	242	81	41	19	19	yes	no
BIOMD0000000175	120	194	94	6	30	145	yes	no
BIOMD0000000255	504	1654	486	317	17	23	yes	no
BIOMD0000000562	0	0	–	–	–	–	–	–

Table S3 Catalyst conventions in the `BioModels` entries. Mod.: number of `SBML` modifier references, which the reader does not import. δ is the deficiency with species listed on both sides kept (as imported), with those species cancelled as in net-stoichiometry tools, and with every modifier added to both sides of its reaction. The reaction-free control `BIOMD0000000562` is omitted. Input files are those of Table S2, checked by SHA-256.

Model	Mod.	δ kept	δ cancelled	δ with modifiers
`BIOMD0000000001`	0	0	0	0
`BIOMD0000000010`	5	0	0	5
`BIOMD0000000051`	5	8	8	13
`BIOMD0000000064`	10	0	0	0
`BIOMD0000000108`	2	8	8	8
`BIOMD0000000404`	2	13	11	15
`BIOMD0000000637`	0	9	9	9
`BIOMD0000000019`	0	41	41	41
`BIOMD0000000175`	142	6	6	8
`BIOMD0000000255`	0	317	317	317

S5 KEGG: module structure and glycolysis controls

Table S4 summarizes structural deficiency and conserved moiety pools across four KEGG pathway modules: glycolysis (`M00001`), pyruvate oxidation (`M00307`), citrate cycle (`M00009`), and pentose phosphate (`M00004`). Conserved pools isolate nucleotide pairs, quinones, ferredoxins, and coenzyme A without manual curation.

Table S4 Structural results for four reduced KEGG modules. m: species, r: directed reactions, n: complexes, ℓ: linkage classes, s: rank, $\delta = n - \ell - s$: deficiency, P: minimal conserved pools, and S: minimal siphons.

Module	m	r	n	ℓ	s	δ	P	S
`M00001`	14	15	16	4	11	1	3	3
`M00307`	13	8	14	7	6	1	8	6
`M00009`	22	12	23	11	12	0	12	9
`M00004`	12	12	14	4	10	0	2	4

S5.1 Glucose-to-pyruvate pathway realization and invariants

Table S5 lists the reaction multiset converting one glucose to two pyruvates across a preparatory phase (reactions firing once) and payoff phase (firing twice). Aldolase (`R01068`) and triosephosphate isomerase (`R01015`) produce two glyceraldehyde 3-phosphate equivalents, directing payoff enzymes (`R01061`–`R00200`) to fire twice. State-space search yields a certified 15-step sequence (Fig. 5b) preserving the weighted three-carbon backbone ($2 \times$ hexoses $+ 1 \times$ trioses $= 2$).

Table S5 Non-zero firing counts of the prescribed KEGG glycolysis flux. Identifiers match `M00001` reaction records.

Reaction	Count	Reaction	Count
`R01786`	1	`R01061`	2
`R13199`	1	`R01512`	2
`R00756`	1	`R01518`	2
`R01068`	1	`R00658`	2
`R01015`	1	`R00200`	2

S5.2 Generic compound identifiers and cofactor boundary controls

KEGG writes polyphosphate glucokinase (R02189, EC 2.7.1.63) as C00404 + C00267 <=> C00404 + C00668. The identifier C00404 names a polyphosphate chain of any length, so $(\text{Phosphate})_n$ on the left and $(\text{Phosphate})_{n-1}$ on the right share it and the polymer appears on both sides. Counting phosphorus shows that one phosphate moves from the polymer to glucose: polyphosphate is a phosphoryl donor, not a retained catalyst. SynKit keeps the equation as written, so in the reduced glycolysis model the polymer is a species that this reaction never consumes. We therefore do not use R02189 as an example of participant retention; the main text uses the catalysts of BIOMD0000000404 instead (Table S3).

Table S6 tests the prescribed reaction multiset from one glucose token and zero tokens of every other species. Removing currency metabolites permits the 15-step carbon-backbone certificate (14 species). Retaining them (24 species) leaves hexokinase without ATP, and no reaction of the requested multiset can fire from the initial marking. The negative result concerns this flux and inventory.

Table S6 Currency-boundary control for the prescribed full glycolytic flux from one glucose and zero tokens of all other species. The negative decision is exhaustive for this reaction multiset.

Currency metabolites	Species	Reactions	Decision	Certificate length
Removed	14	15	realizable	15
Retained	24	15	unrealizable	–

S5.3 Initial cofactor inventories and firing order

Table S7 holds the full prescribed reaction multiset fixed and varies only the initial inventory. From glucose alone (I_0), the adenylate set $\{\text{ATP}, \text{ADP}\}$ is an empty siphon in the selected reaction subnet, which rules out ATP-dependent startup. Supplying ATP alone (I_1, I_2) still leaves net deficits of two ADP, two NAD^+ and two orthophosphate tokens. Adding NAD^+ removes only its deficit (I_3); adding phosphate as well still leaves an ADP deficit (I_4). These obstructions follow already from $m_0 + \mathbf{S}f \not\geq 0$.

Inventory I_5 supplies one ATP, two ADP, two NAD^+ and two orthophosphate tokens alongside glucose. Its predicted final inventory is non-negative, but the prescribed multiset is unrealizable: two ATP-consuming preparatory steps must occur before any ATP-producing payoff reaction. Exhaustive search stops after three markings. With two initial ATP instead (I_6), an independently replayed 15-step certificate completes the same multiset after exploring 30 markings. The final inventory is two pyruvates, four ATP, two NADH, two water and two protons, with no unlisted tokens. No case receives replenishment during firing. This paired control distinguishes aggregate stoichiometric sufficiency from an enabled firing order.

S6 Formose networks: expansion and precursor requirements

The formose reaction system serves as a benchmark for rule-based graph transformation, exact mass conservation, and discrete pathway realizability in open-ended reaction spaces. Graph transformation explores

Table S7 Initial-inventory controls for the prescribed full glycolytic flux. Each case starts with one glucose and the listed cofactor tokens, with zero tokens of all other species. P_i denotes orthophosphate; Net tests $m_0 + \mathbf{S}f \geq 0$, States counts explored markings, and Length is the verified certificate length. Negative decisions are exhaustive for the prescribed multiset.

Case	ATP	ADP	NAD^+	P_i	Net	Decision	States	Length
I_0	0	0	0	0	no	unrealizable	1	–
I_1	1	0	0	0	no	unrealizable	3	–
I_2	2	0	0	0	no	unrealizable	6	–
I_3	2	0	2	0	no	unrealizable	6	–
I_4	2	0	2	2	no	unrealizable	27	–
I_5	1	2	2	2	yes	unrealizable	3	–
I_6	2	2	2	2	yes	realizable	30	15

carbohydrate condensation from formaldehyde (C_1) and glycolaldehyde (C_2) seeds through recursive aldol coupling and enediol-mediated keto-enol tautomerization.

S6.1 Network growth and stoichiometric carbon conservation

Table S8 reports network growth from the formaldehyde and glycolaldehyde seeds. After four iterations, the network contains 37 species and 44 reactions and reaches species with ten carbon atoms (C_{10}). Exact nullspace analysis of its stoichiometric matrix identifies a single positive conservation semiflow whose weights match the carbon atom count of every species, confirming carbon conservation across the generated reactions.

S6.2 Comparative rule expansion benchmarking

Table S8 compares the full API runtime of `SynReactor` DPO against `MØD` and `MolFoundry` across six expansion depths from fresh formaldehyde and glycolaldehyde seeds. Timings are evaluated across 20 timed repetitions after three warm-ups with all tool processes bound to one logical CPU. Each tool uses its native rule format, and generated molecules are canonicalized without explicit hydrogens to evaluate exact network equivalence. Across all completed depths, all three tools generate identical species sets and directed stoichiometric reactions, validating algorithmic equivalence.

Table S8 Full API runtime comparison. Times are median cumulative seconds from seeds over 20 timed runs after three warm-ups, rounded to three significant figures as in the main text. Native `SynReactor` uses its C++ controller and DPO kernel with implicit hydrogen counts, whereas `MØD` and `MolFoundry` use explicit hydrogen vertices. Rule preparation and network construction are included for every tool. Counts describe identical completed species and stoichiometric reaction sets. Each expansion has a 180 s time limit and an 8 GiB per-process address-space limit.

Depth	Species	Reactions	`SynReactor` C++ DPO	`MØD`	`MolFoundry`
1	3	1	0.0275	0.000705	0.0031
2	5	4	0.0326	0.000947	0.00634
3	9	10	0.0272	0.00152	0.00793
4	37	44	0.0283	0.00369	0.0229
5	302	371	0.0642	0.0362	0.26
6	10572	11239	1.86	2.22	15.1

S6.3 Route definitions and finite precursor inventories

The targets are an aldotetrose (T_4, `O=CC(O)C(O)CO`) and an aldohexose (T_6, `O=CC(O)C(O)C(O)C(O)CO`). Under qualitative set reachability, both are reachable from the two seed species. Table S9 distinguishes qualitative depth from the reaction multiplicities required for discrete firing, and Fig. S4 shows the reaction motifs.

Table S9 Qualitative depths and prescribed discrete reaction multisets for the selected formose targets. Set reachability reuses a species once generated, whereas discrete firing requires a separate token for each consumed molecule.

Target	Qualitative depth	Discrete firings	Prescribed multiset
T_4	2	2	$r_1 + r_3$
T_6	3	4	$2r_1 + r_3 + r_7$

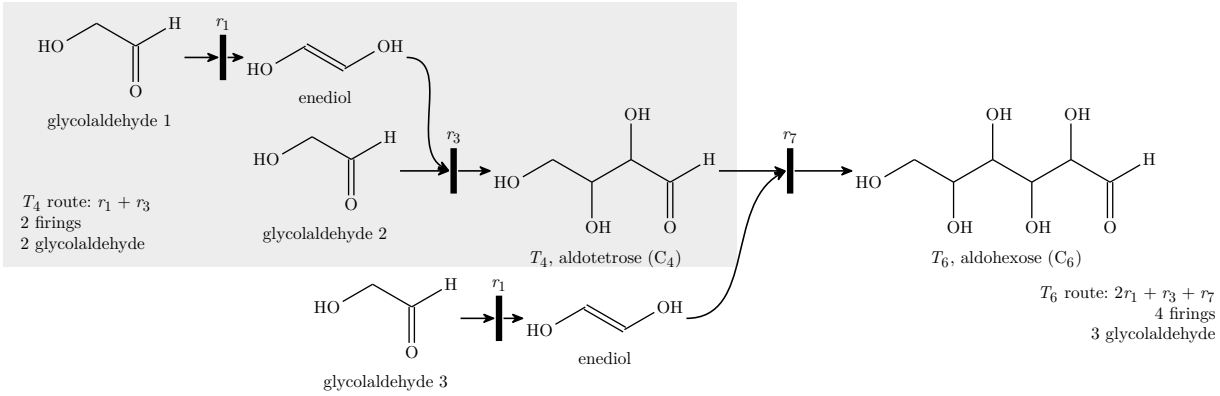


Figure S4 Precursor requirements of the selected formose routes. Bars are reactions, as in the Petri nets of the main text, and each glycolaldehyde is drawn as a separate molecule because discrete firing consumes one token per molecule. Keto–enol tautomerization (r_1) supplies the enediol consumed by each aldol addition (r_3, r_7). The shaded block is the T_4 route ($r_1 + r_3$: two firings, two glycolaldehyde), and the whole scheme is the T_6 route ($2r_1 + r_3 + r_7$: four firings, three glycolaldehyde). Structures are constitutional, with no stereochemistry or atom mapping assigned.

Table S10 evaluates five initial precursor inventories. Formaldehyde is C_1 and glycolaldehyde is C_2. Every positive sequence is confirmed by a replayable firing sequence, while negative decisions are established by exhaustive search over the prescribed multiset. For an independent check, the same reaction multisets and initial markings are supplied to `MolFoundry`'s `FlowSolution`, without optimizing a new hyperflow. Its `is_realizable()` method and SMT certificate enumeration agree with all ten `SynKit` decisions. All three positive certificates are independently replayed, checking enabling before every firing, the requested reaction multiplicities, and target production. This comparison concerns the prescribed routes, rather than unrestricted target reachability. Both prescribed routes consume no formaldehyde: their different outcomes among the equal-carbon inventories follow from glycolaldehyde counts.

Table S10 Realizability of the prescribed tetrose and hexose routes across five precursor inventories. Counts denote molecules and total carbon counts carbon tokens. Yes indicates a verified certificate and No an exhaustive negative decision for the selected multiset.

Pool	C_1 count	C_2 count	Total carbon	T_4	T_6
M_1	1	1	3	no	no
M_2	2	1	4	no	no
M_3	4	1	6	no	no
M_4	2	2	6	yes	no
M_5	0	3	6	yes	yes

The tetrose route converts glycolaldehyde to enediol (r_1) and couples that intermediate with a second glycolaldehyde (r_3). This motif requires two glycolaldehyde precursors and consumes no formaldehyde, so inventories M_4 and M_5 support it while M_3 (containing only one glycolaldehyde) cannot. The hexose route requires two successive aldol couplings consuming two enediol equivalents, demanding three glycolaldehyde

precursors in total ($2r_1 + r_3 + r_7$), which only pool M_5 satisfies. Consequently, the selected multisets have different realizability outcomes in the three six-carbon inventories (M_3, M_4, and M_5). The negative decisions do not exclude alternative target-producing multisets from these mixed inventories.

An independent structural obstruction applies to a formaldehyde-only inventory: one eight-species minimal siphon of the generated network contains glycolaldehyde (`O=CCO`), its enediol (`OC=CO`), and both targets, but excludes formaldehyde (`C=O`). If all eight species are absent initially, no reaction can produce a member, so formaldehyde alone cannot initiate either target. Each inventory M_1–M_5 already marks this siphon through glycolaldehyde; their selected-route decisions therefore require the separate multiplicity analysis above.

References

[1] Martin Feinberg. *Foundations of Chemical Reaction Network Theory*, volume 202 of *Applied Mathematical Sciences*. Springer, Cham, 2019. doi:10.1007/978-3-030-03858-8.

[2] Boris B Edelstein. Biochemical model with multiple steady states and hysteresis. *Journal of Theoretical Biology*, 29(1):57–62, 1970. doi:10.1016/0022-5193(70)90118-9.

[3] Liming Wang and Eduardo D Sontag. On the number of steady states in a multiple futile cycle. *Journal of Mathematical Biology*, 57(1):29–52, 2008. doi:10.1007/s00285-007-0145-z.

[4] Guy Shinar and Martin Feinberg. Structural sources of robustness in biochemical reaction networks. *Science*, 327(5971):1389–1391, 2010. doi:10.1126/science.1183372.